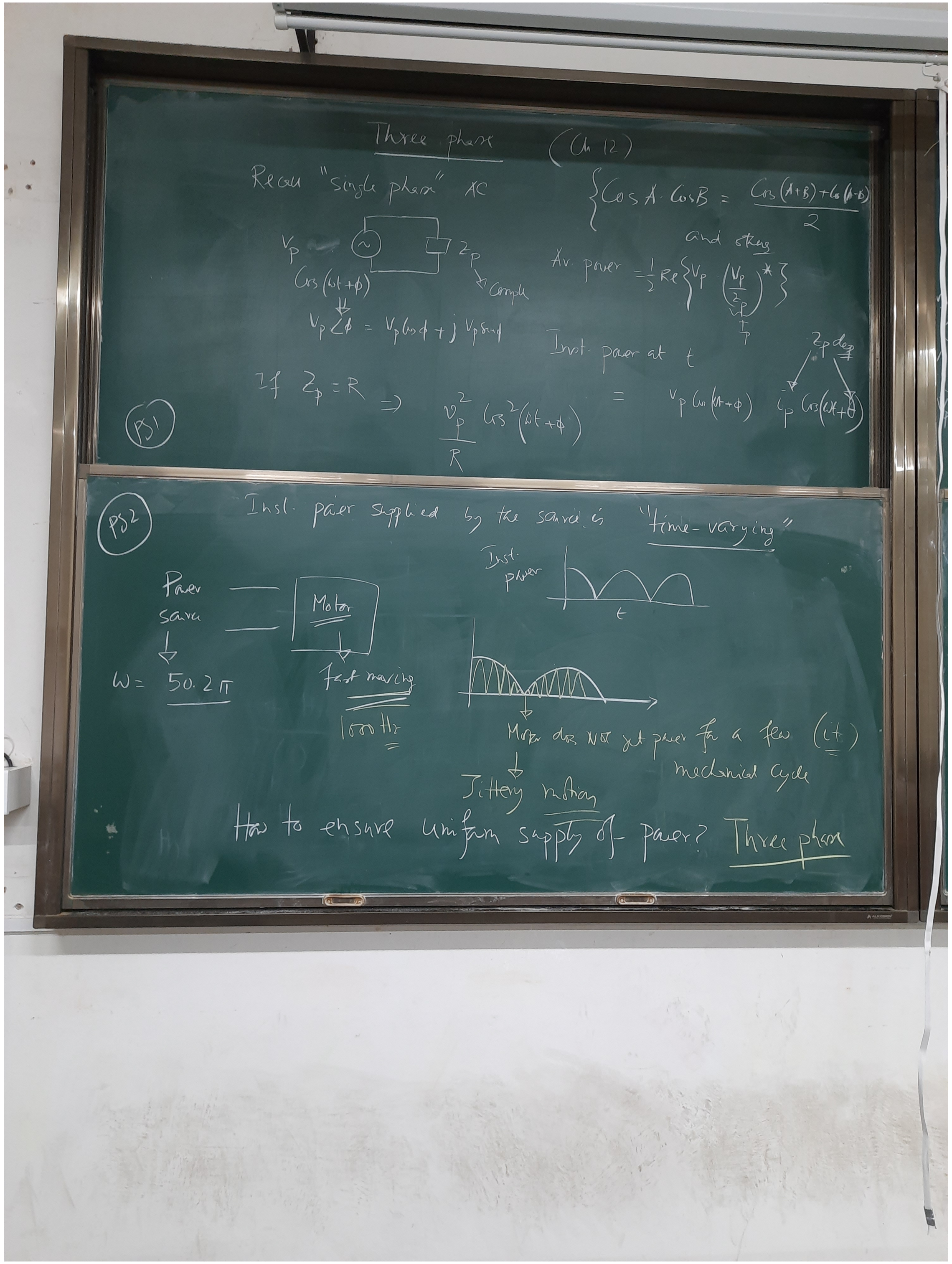


Three phase (Ch 12)

Recall "single phase" AC

$V_p \cos(\omega t + \phi)$ → Z_p → Complex

$\Downarrow$

$V_p \angle \phi = V_p \cos\phi + j V_p \sin\phi$

If $Z_p = R \Rightarrow \dfrac{V_p^2}{R} \cos^2(\omega t + \phi)$

PS1

$\left\{ \cos A \cdot \cos B = \dfrac{\cos(A+B) + \cos(A-B)}{2} \right.$

and others

Av. power $= \frac{1}{2} \mathrm{Re} \left\{ V_p \cdot \left(\dfrac{V_p}{Z_p} \right)^* \right\}$ ($\dfrac{V_p}{Z_p} \to I_p$)

Inst. power at t

$= V_p \cos(\omega t + \phi) \; i_p \cos(\omega t + \theta)$ (Z_p deg)

PS2

Inst. power supplied by the source is "time-varying"

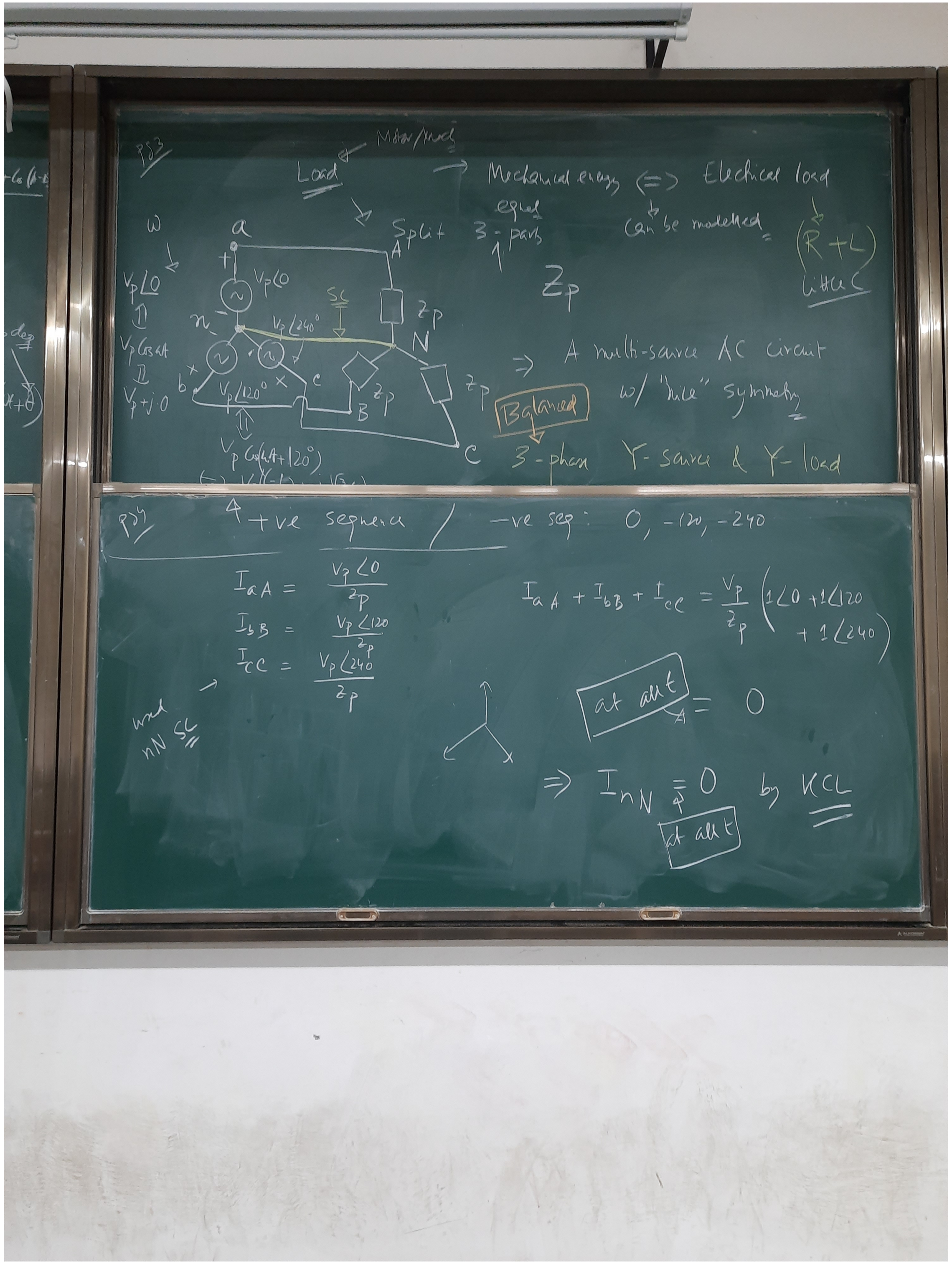
Power source ↓ $\omega = 50 \cdot 2\pi$

Motor ↓ Fast moving 1000 Hz

↓ Motor does NOT get power for a few (i.t.) mechanical cycle

↓ Jittery motion

How to ensure uniform supply of power? Three phase



p53

Load ← Motor/mach

→ Mechanical energy $\Leftrightarrow$ Electrical load

Can be modelled = (R + L) little C

↘ Split 3 - parts (equal)

Z_p

ω

$V_p\angle 0$

$\Updownarrow$

$V_p \cos\omega t$

$\Updownarrow$

$V_p + j\cdot 0$

$V_p \cos(\omega t + 120^\circ)$

⇒ A multi-source AC circuit w/ "nice" symmetry

Balanced

3 - phase Y - source & Y - load

p54

+ve sequence / −ve seq: 0, −120, −240

$$I_{aA} = \frac{V_p\angle 0}{Z_p}$$

$$I_{bB} = \frac{V_p\angle 120}{Z_p}$$

$$I_{cC} = \frac{V_p\angle 240}{Z_p}$$

used nN SC

$$I_{aA} + I_{bB} + I_{cC} = \frac{V_p}{Z_p}\left(1\angle 0 + 1\angle 120 + 1\angle 240\right)$$

$\boxed{\text{at all } t}$ = 0

$\Rightarrow I_{nN} \overset{?}{=} 0$ by KCL

$\boxed{\text{at all } t}$

(P35)

What if aA, bB, cC or nN "wires" have impedance?

easy ⇒ gets captured in Z_p

as long as $Z_{aA} = Z_{bB} = Z_{cC}$

$I_{nN} = 0$ & prev I_{aA}, I_{bB}, I_{cC}

is still a soln for any $Z_{nN} \neq 0$

⇓

Match KVL / KCL ← Indeed THE soln

Since KVL/KCL lin circ ⇒ unique soln

(P36)

Inst. power at t =

$V_p \cos(\omega t + \theta)$

$Z_p \to |Z_p|(\cos\phi + j\sin\phi)$ ← in Z_p

0/120/240

$\Rightarrow i_p \cos(\omega t + \theta - \phi)$

$i_p = \frac{V_p}{|Z_p|}$

Recall $\cos A \cos B = \frac{\cos(A+B) + \cos(A-B)}{2}$

$P_{aA}(t) + P_{bB}(t) + P_{cC}(t)$

$$= v_p i_p \left(\sum_{\theta} \cos(\omega t + \theta) \cos(\omega t + \theta - \phi) \right)$$

θ ↓ 0/120/240

$$= v_p i_p \sum_{\theta} \frac{1}{2}\left(\cos\phi + \cos(2\omega t + 2\theta - \phi) \right)$$

← indep of t

$$\cos(2\omega t-\phi+0)+\cos(2\omega t-\phi+240^\circ)+(\qquad +480^\circ)$$

$$=0$$

Ch 12

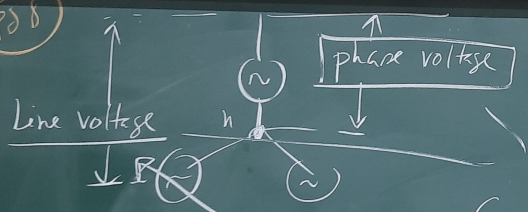
Hence $p_{aA}(t)+p_{bB}(t)+p_{cC}(t)$ = SAME at all t

Avg. power consumed $\propto$ → only depends on $\underline{\phi}$ ↓ $\cos\underline{\phi}$

(get the algebra) ↓ both

PS7

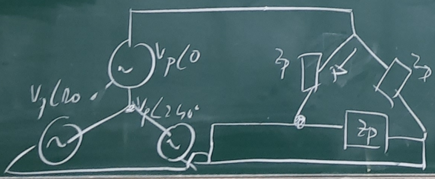
PS8



simple trig. to relate them

Can write av power in terms of

Balanced Y- source & Δ- load



Inst. power SAME at all t